



Final Exam

Duration: 1h 30min

➡ **Answers must be numbered, clear, and justified.**

--- **Exercise 1** (4 points) -----

1. Indicate, with justification, if the following statement is true or false, then give its negation.

$$\exists x \in \mathbb{R}, x^2 - 5 \geq 0.$$

2. Using the proof by contrapositive, prove that $\forall m \in \mathbb{Z}$:

$$m^2 - 3 \text{ is odd} \Rightarrow m \text{ is even.}$$

--- **Exercise 2** (7 points) -----

I. Let $\mathcal{R}$ be a binary relation defined on $\mathbb{Z}$ as follows:

$$a \mathcal{R} b \Leftrightarrow \exists n \in \mathbb{N}, a - b = 3n.$$

1. Is $(-6)\mathcal{R}(-3)$? , $(-3)\mathcal{R}(-6)$? Justify.

2. Show that $\mathcal{R}$ is an order relation.

3. Is the order total? Justify.

II. Let f be a function defined as follows:

$$f : [-1, 3] \rightarrow [0, 2]$$

$$x \mapsto \frac{x^2}{x+2}$$

1. Determine $f(\{-1, 0, 2\})$ and $f^{-1}(\{\frac{1}{3}, 2\})$.

2. Is f injective? Surjective? Bijective? Justify.

--- **Exercise 3** (4 points) -----

Let $k = \frac{2i+2\sqrt{3}}{4i}$ be a complex number.

1. Write k in algebraic form $(x + iy)$.

2. Determine the modulus r and an argument θ in $[0, 2\pi]$ for k , then write k in exponential form.

3. Using De Moivre's formula, find the real number k^3 .

4. Solve in $\mathbb{C}$ the equation $z^3 - k^3 = 0$.

--- **Exercise 4** (5 points) -----

I. Let $E = \{(x, y, z) \in \mathbb{R}^3, \frac{1}{2}xy = z\}$ be a subset of $\mathbb{R}^3$.

Let $u = (1, 4, 2)$ and $v = (3, -2, -3)$ be two vectors from $\mathbb{R}^3$.

1. Is $u \in E$? $v \in E$? $u + v \in E$? Justify.

2. Is E a subspace of $\mathbb{R}^3$? Justify.

II. Let $F = \{(x, y) \in \mathbb{R}^2, x - 2y = 0\}$ be a subset of $\mathbb{R}^2$.

Prove that F is a subspace of $\mathbb{R}^2$.

----- Good luck -----